

SyncGuard: Contrastive Audio-Visual Deepfake Detection via Temporal Phoneme-Face Coherence

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INTRODUCTION

The proliferation of generative AI has made photorealistic deepfake videos trivially producible, threatening the integrity of video evidence across journalism, legal proceedings, and political discourse. Face-swap and voice-cloning techniques, once the exclusive domain of well-resourced actors, are now accessible via commodity hardware and open-source toolkits, dramatically lowering the barrier to information warfare and identity fraud. Existing detection systems rely predominantly on visual-only artifact classifiers that achieve impressive accuracy on known forgery pipelines but **fail catastrophically on unseen manipulation techniques**, dropping 20–30 percentage points across dataset boundaries [6]. This brittleness renders them unsuitable for real-world deployment, where adversaries continuously evolve their methods.

SyncGuard is grounded in a physiological observation: authentic human speech is governed by a **tight biomechanical coupling** between articulatory lip movements and acoustic output, a coordination pattern shaped by years of neuromuscular learning. Deepfake methods must either precisely preserve this synchrony or violate it detectably. We exploit this asymmetry as a universal, manipulation-agnostic detection signal, combining frame-level sync-scores with a novel cross-modal attention mechanism that captures identity-level mismatches invisible to scalar similarity measures.

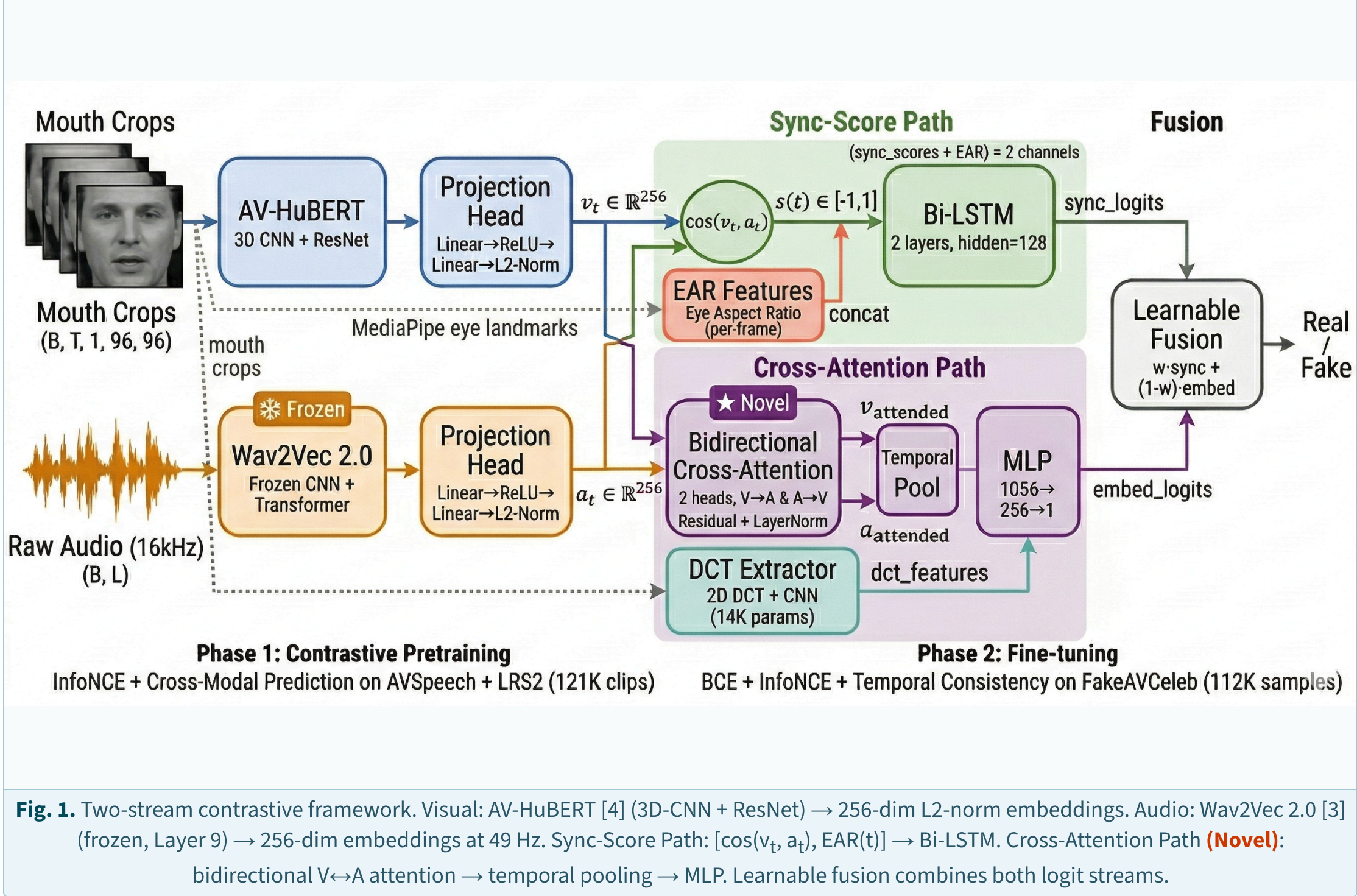
PROBLEM STATEMENT

Despite significant progress in unimodal forgery detection, three fundamental gaps remain unsolved in the audio-visual deepfake detection literature:

- Generalization failure.** State-of-the-art visual detectors collapse to near-chance performance when evaluated on unseen forgery pipelines or different datasets, rendering closed-world accuracy metrics misleading indicators of real-world utility.
- Scalar synchrony is insufficient.** Existing AV detectors reduce multi-dimensional embedding correspondence to a single cosine similarity score, discarding the rich cross-modal structure needed to expose identity-swap attacks. A face-swap that uses the victim’s own voice can maintain high cosine similarity while being clearly detectable in the full embedding space.
- Single-modality blindness.** No deployed system jointly reasons about both audio manipulations (voice cloning) and visual manipulations (face swaps) in a unified framework, leaving hybrid attacks, which fool each detector independently, without a principled defense.

APPROACH / SYSTEM DESIGN

System Architecture

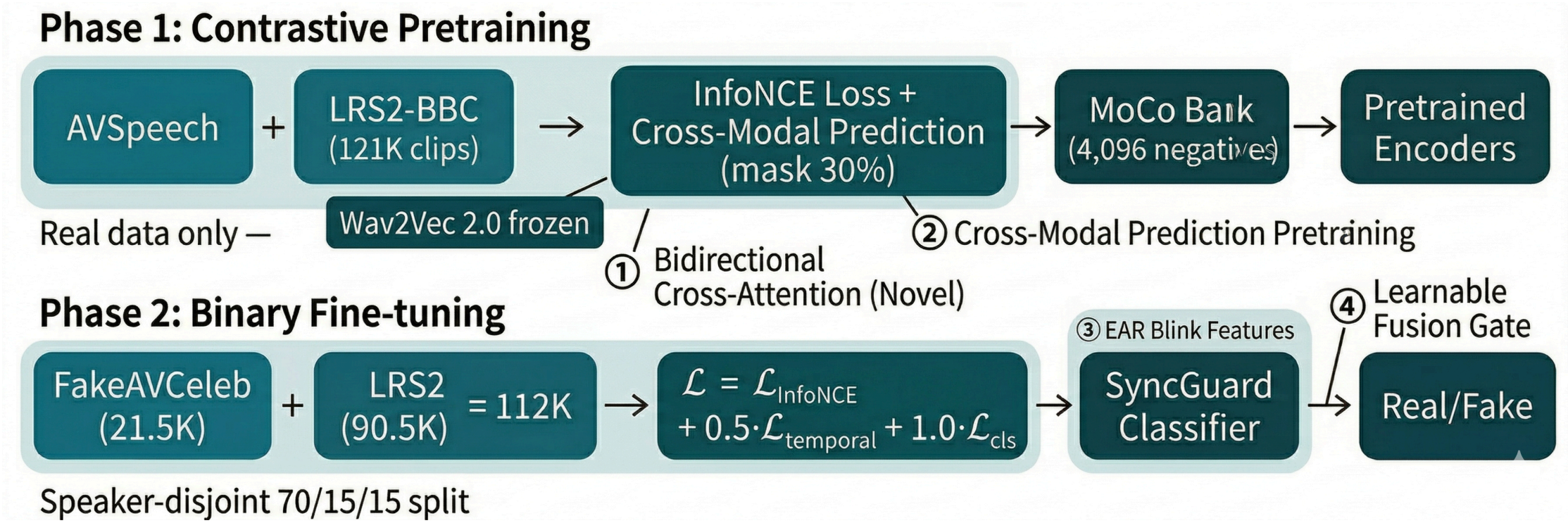


Datasets

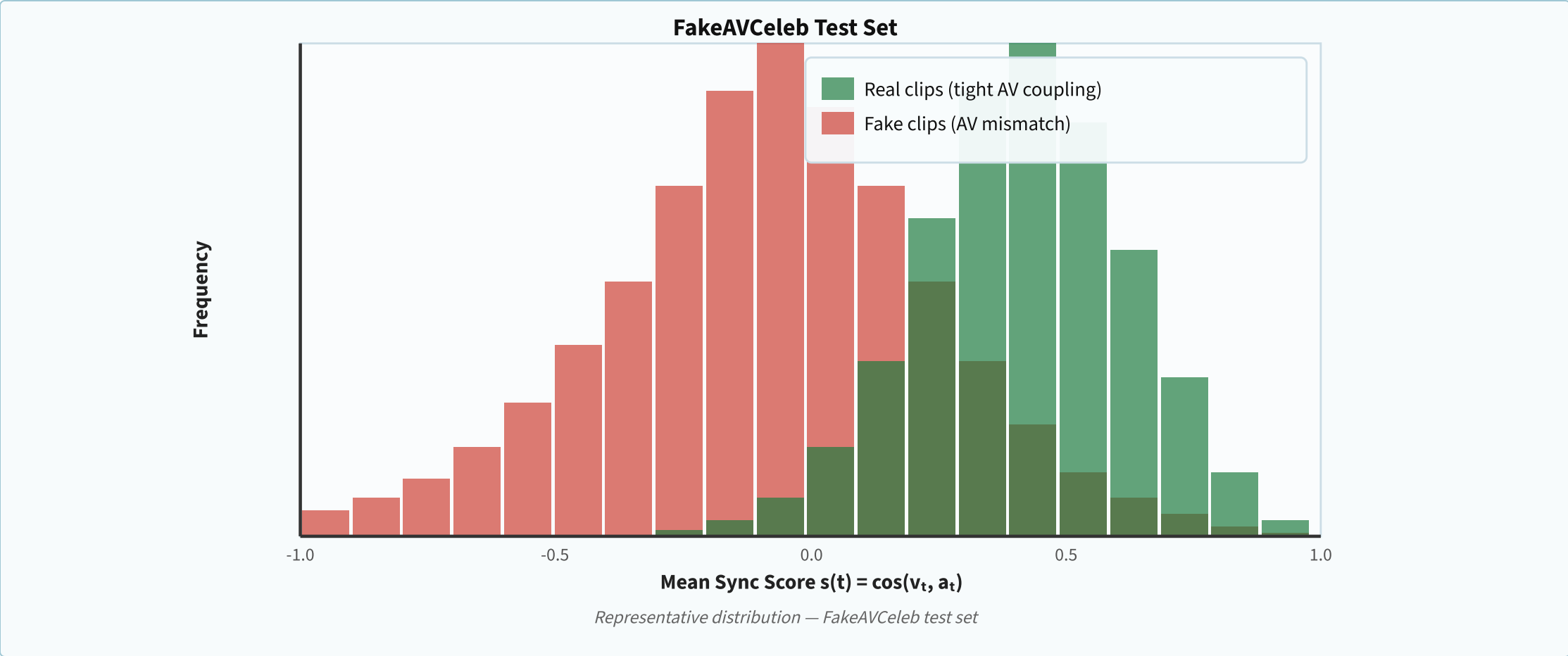
Dataset	Samples	Role
FakeAVCeleb [1]	21,544	Train / Val / Test (4 categories)
AVSpeech	24,760	Contrastive Pretraining
LRS2-BBC	96,318	Pretraining + Real Anchors
DFDC Part 0	1,343	Zero-Shot Cross-Dataset Eval

FakeAVCeleb categories: RV-RA (real), FV-RA (face-swap only), RV-FA (voice-clone only), FV-FA (both). Speaker-disjoint 70/15/15 split.

Training Protocol



Sync-Score Distribution: Real vs. Fake

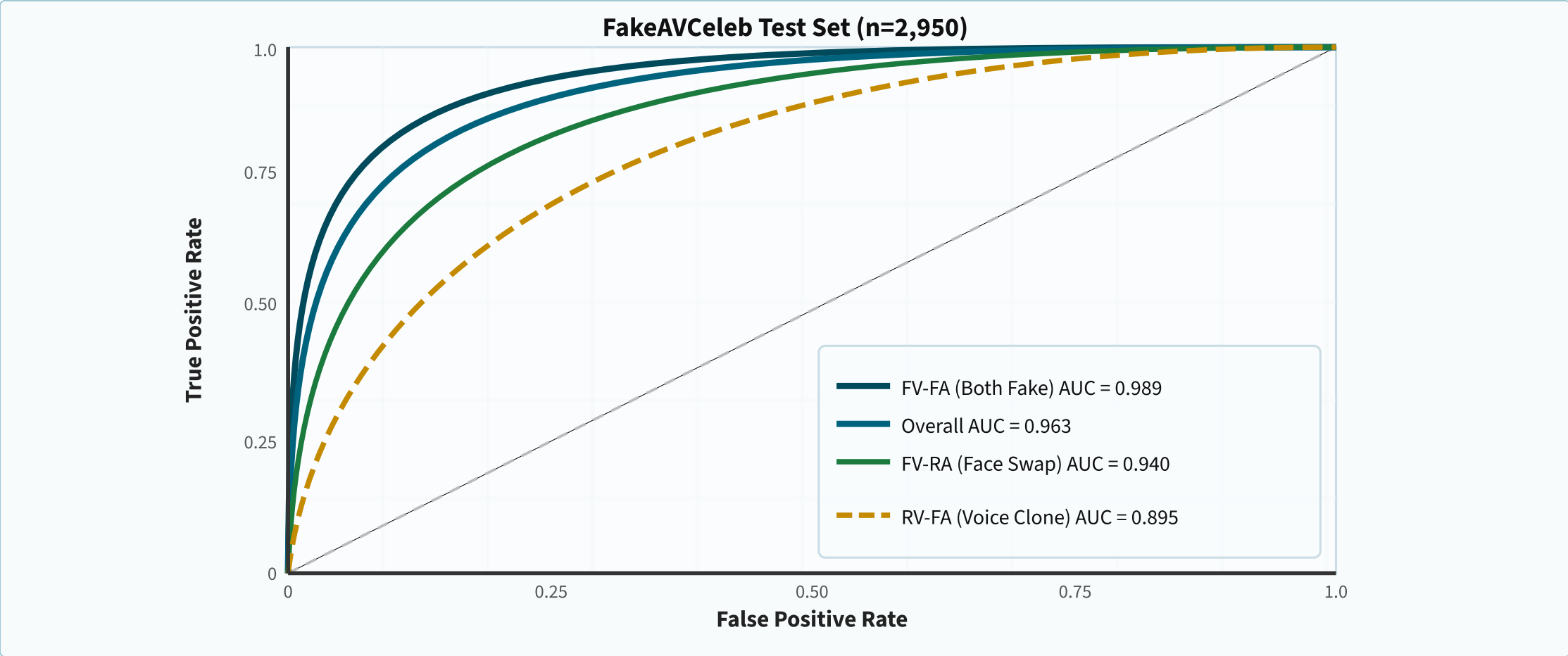


RESULTS

Key Metrics

96.3% AUC-ROC	8.2% EER	98.9% FV-FA AUC	+22.8pp RV-FA GAIN
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ROC Analysis: FakeAVCeleb



Per-Category Breakdown

Category	Description	AUC	EER
RV-RA	Real video, real audio	N/A	N/A
FV-RA	Face-swap, real audio	0.940	0.103
RV-FA (Sync-Only)	Voice-cloned, no cross-attn	0.667	0.284
RV-FA	Voice-cloned audio	0.895	0.118
FV-FA	Both modalities fake	0.989	0.034

Cross-Dataset & SOTA Comparison

Zero-Shot Ablation: DFDC Part 0 (n=1,343)		
Model Variant	AUC	EER
Sync-Score Only	0.458	0.512
+ Cross-Attn (Ours)	0.526	0.491

Δ DFDC face-swaps preserve lip-sync, an open problem across the field.

Published Methods: FakeAVCeleb		
Method	Venue	AUC
AVoiD-DF [8]	IEEE TIFS '23	89.2%
MRDF-CE [9]	ICASSP '24	92.4%*
SyncGuard (Ours)	2026	96.3%
AVFF [2]	CVPR '24	99.1%†

† Benchmark bias [7]. *Classification accuracy, not AUC.

CONCLUSIONS & FUTURE WORK

What We Demonstrated

- Competitive contrastive AV detection:** 96.3% AUC / 8.2% EER on FakeAVCeleb, surpassing AVoiD-DF [8] (89.2%) and MRDF-CE [9] (92.4%), approaching AVFF (99.1%†, likely inflated by benchmark bias [7]).
- Cross-attention closes the voice-clone gap:** +22.8pp on RV-FA (66.7% $\rightarrow$ 89.5%); explicit cross-modal attention captures spectral-visual mismatches invisible to frame-level synchrony alone. This is the single largest per-category improvement in our system, demonstrating that identity-coherence reasoning, not just temporal alignment, is essential for detecting modern neural voice-cloning attacks.
- Dual-path fusion generalizes:** Learnable fusion weight w balances both paths across face-swap (0.940), voice-clone (0.895), and fully synthetic (0.989) attacks without manipulation-type supervision.

Limitations

- Synchrony fails against lip-sync-preserving face swaps:** DFDC zero-shot yields 52.6% AUC (near chance) because methods like SimSwap retain accurate lip motion, nullifying temporal coherence signals.
- FakeAVCeleb leading-silence bias:** Forged clips frequently begin with silence artifacts absent in real clips. Our 96.3% AUC may overestimate true discrimination ability until a debiased benchmark is available.

Future Directions

- Self-Blended Image augmentation:** Applying SBI [5] during fine-tuning would expose the model to blending-boundary artifacts, directly targeting the synchrony-blind failure mode on DFDC without additional labeled data.
- Foundation model backbones (CLIP, DINOv2):** Replacing AV-HuBERT’s visual encoder with a ViT pretrained on internet-scale data would provide identity-level features sensitive to face-swap artifacts independent of lip motion.

REFERENCES

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